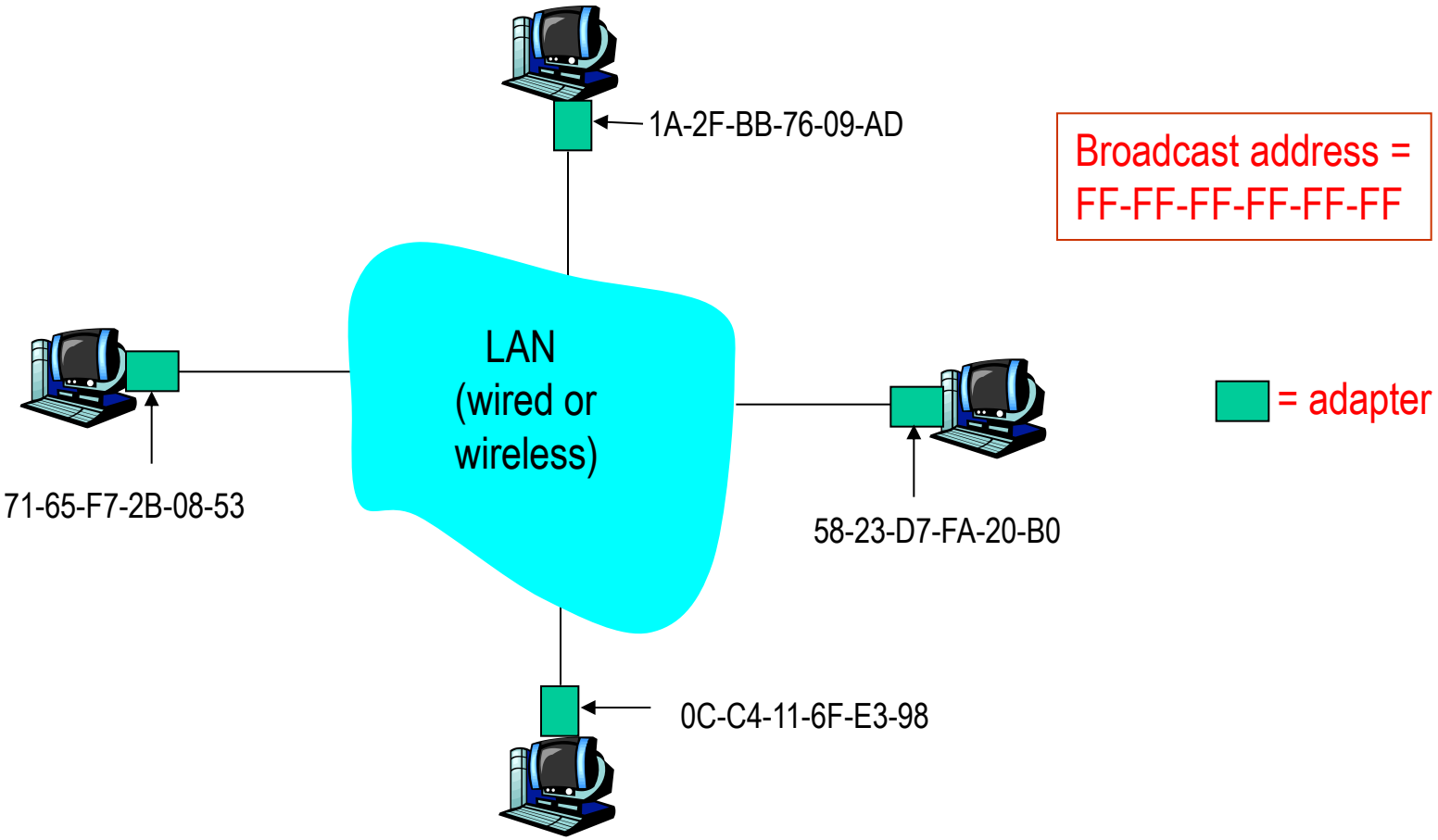

Computer Networks
Lecture 13
Shmulik Kaplan

Today

- Recap
- DHCP – some final details
- LAN technologies & collisions
  - T – connector bus
  - Hub
  - Bridge & Switch
- A life of a web request
- Questions

LAN Addresses and ARP

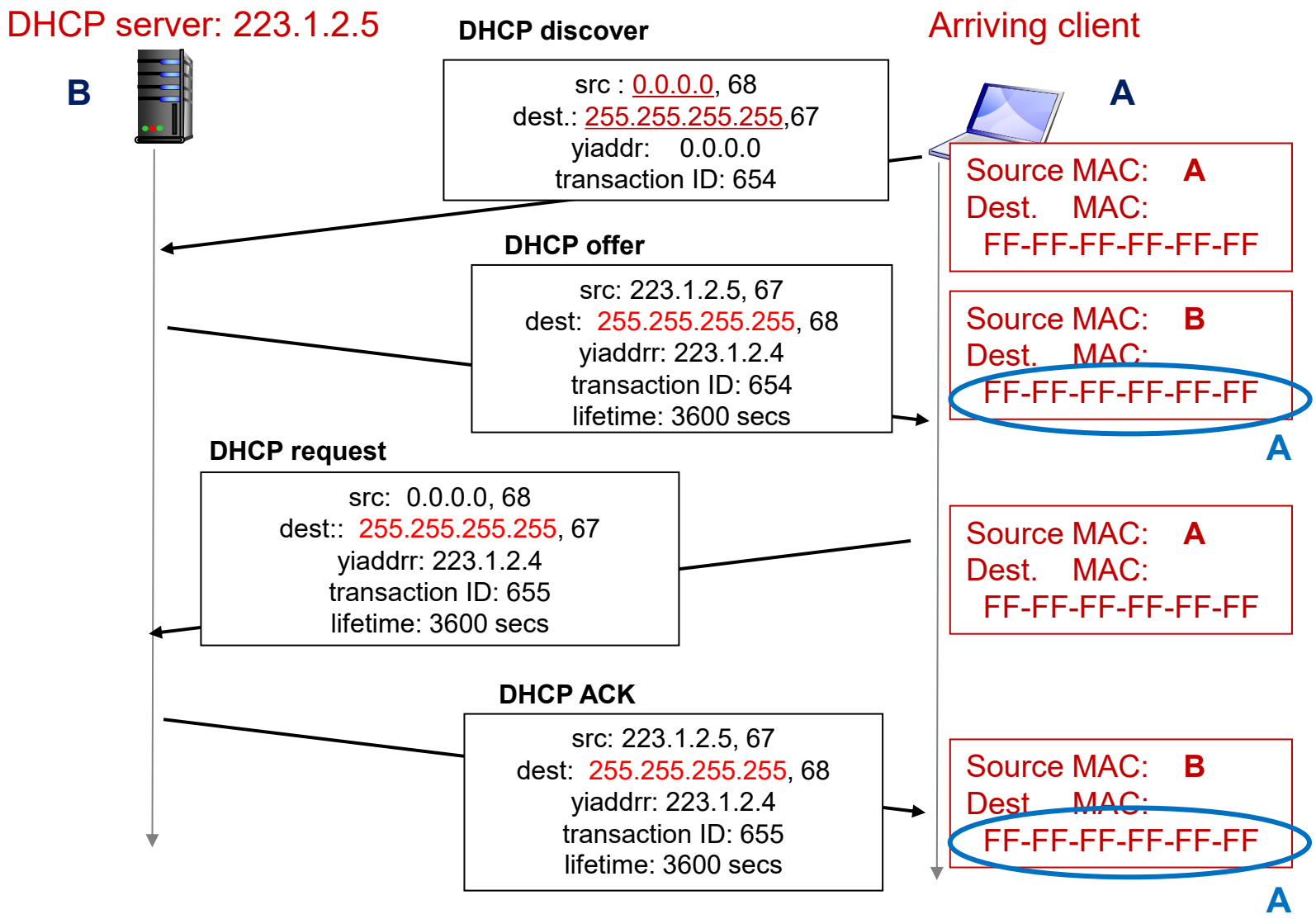
Each adapter on LAN has unique LAN address



Today

- ✔ Recap
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- Questions

DHCP client-server scenario



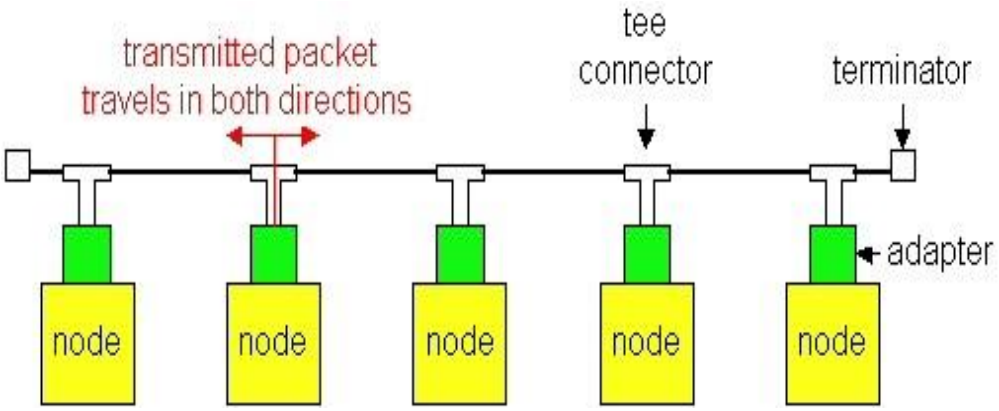
The client can use a broadcast flag to indicate in which way (broadcast or unicast) it should receive the DHCP reply from the server (RFC 2131). The default is unicast.

Today

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- ✔ DHCP – some final details
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Ethernet Technologies: 10Base2

- 10: 10Mbps ; 2: under 200 meters max cable length
- thin coaxial cable in a bus topology







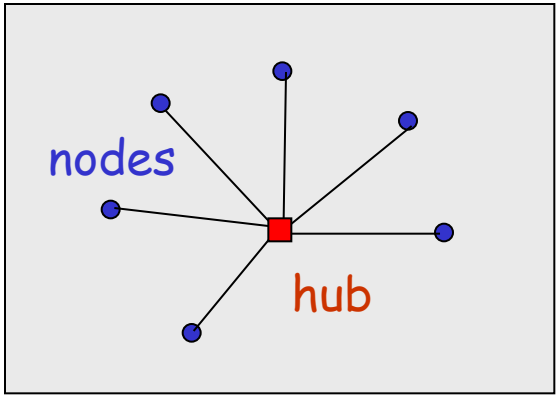
- **repeaters** used to connect up to multiple segments
- repeater repeats bits it hears on one interface to its other interfaces: physical layer device only!



Same collision domain!

10BaseT and 100BaseT

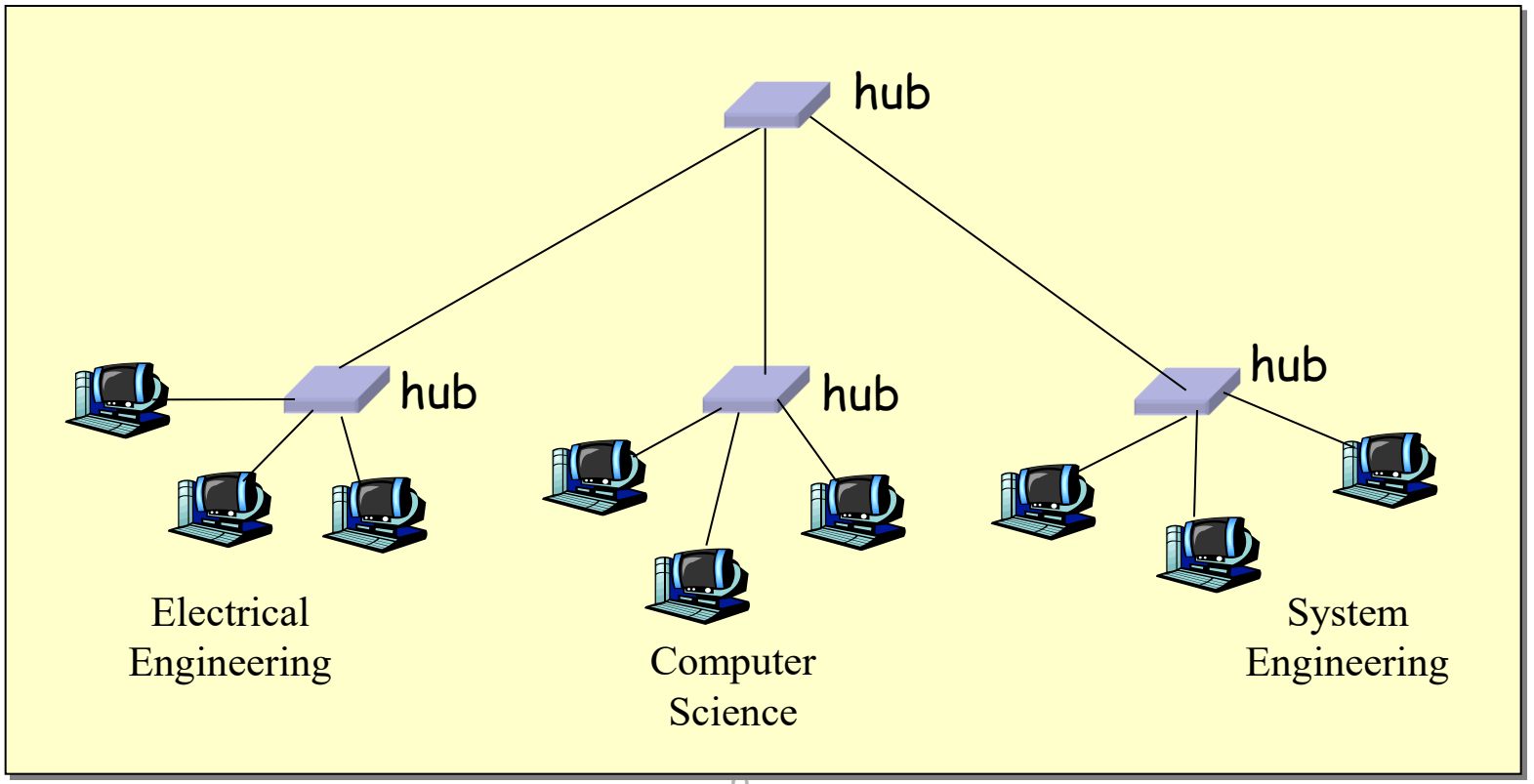
- 10/100 Mbps rate; latter called “fast ethernet”
- T stands for Twisted Pair
- Different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1Gbps, 10 Gbps, 40 Gbps …
- Nodes connect to a hub: “star topology”; 100m max distance between nodes and hub



- Hubs are essentially physical-layer repeaters:
  - Bits coming in one link go out all other links
  - A plain hub doesn't have its own MAC address or IP address
  - However, a managed hub … provides network management functionality

Interconnecting with hubs

- Backbone hub interconnects LAN segments
- Extends max distance between nodes
- If a node in CS and a node EE transmit at same time: collision
- Can't interconnect 10BaseT & 100BaseT



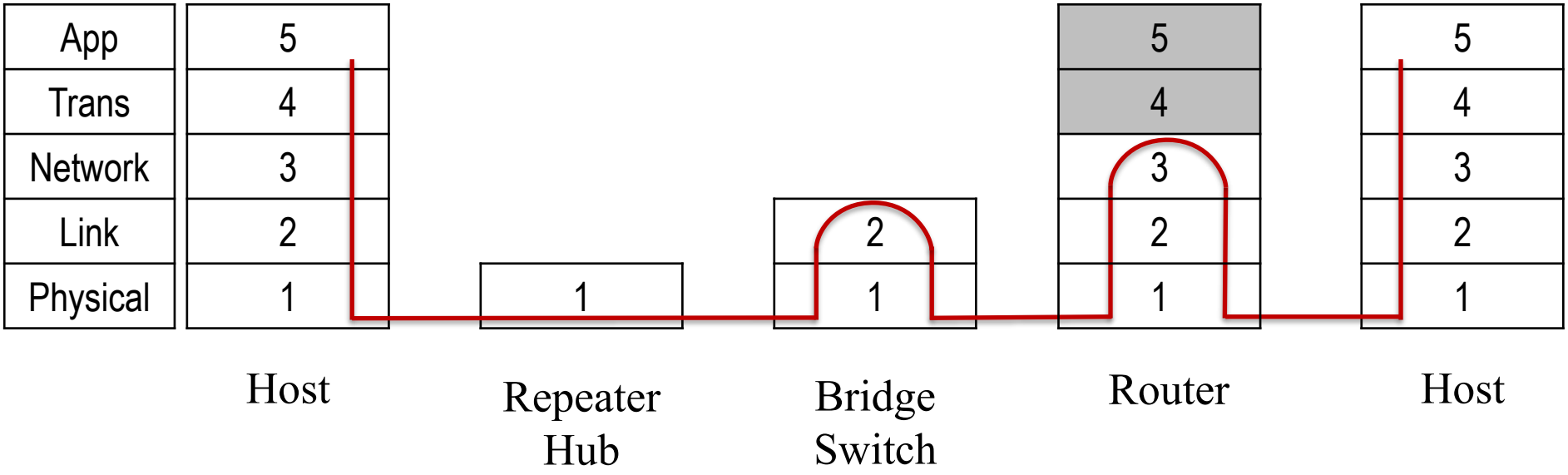
Today

- ✓ Recap
- ✓ DHCP – some final details
- LAN technologies & collisions
  - ✓ T – connector bus
  - ✓ Hub
    - Bridge & Switch
- A life of a web request
- Questions

Interconnecting LAN segments

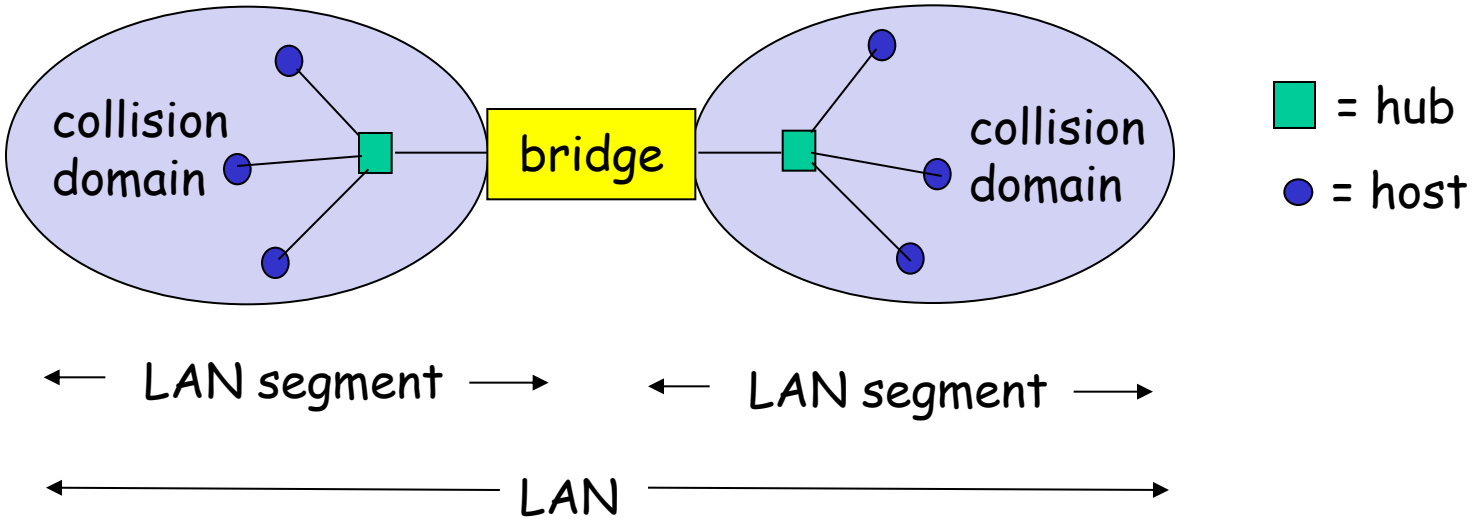
- Hubs
- Bridges
- Switches
  - Remark: switches are essentially multi-port bridges.
  - What we say about bridges also holds for switches!

Communication Devices



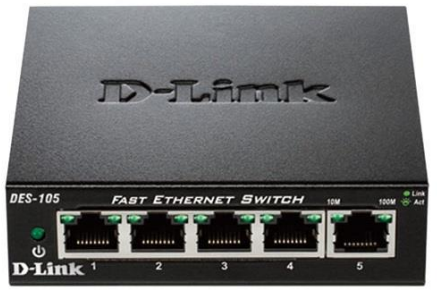
Bridges/switch: traffic isolation

- Bridge installation breaks LAN into LAN segments
- bridges filter packets:
  - same-LAN-segment frames not usually forwarded onto other LAN segments
  - segments become separate collision domains



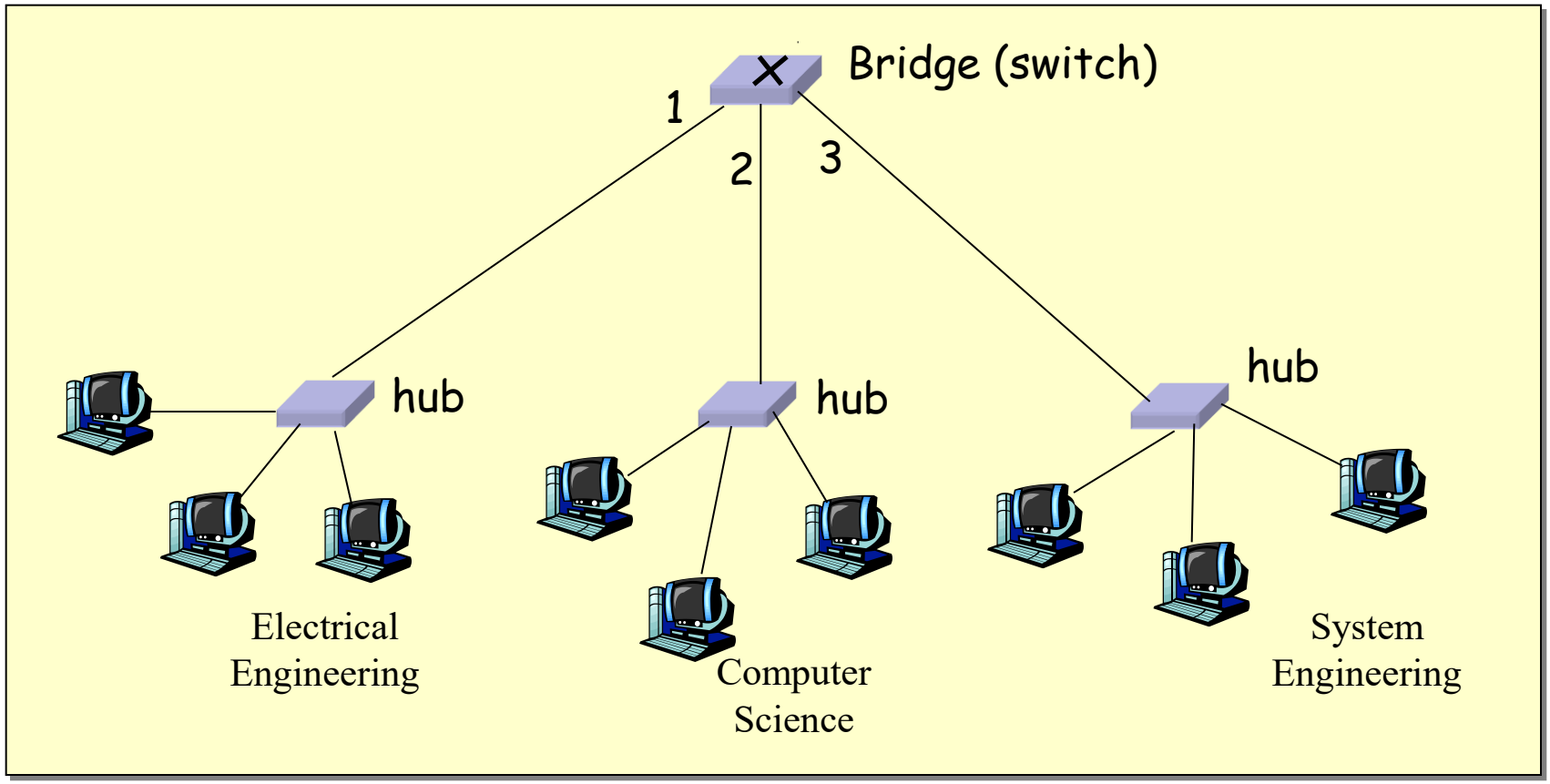


Bridges/Switches



- Link layer device
  - stores and forwards Ethernet frames
  - examines frame header and selectively forwards frame based on MAC destination address
  - when frame is to be forwarded on segment, uses CSMA/CD to access segment
- Transparent
  - hosts are unaware of presence of bridges
- plug-and-play, self-learning
  - bridges do not need to be configured

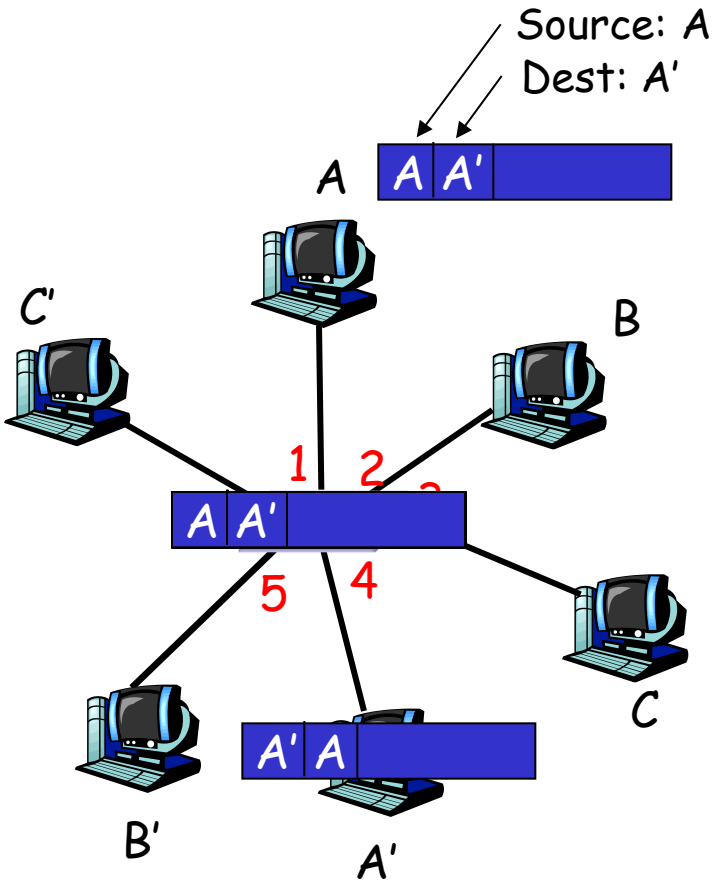
Forwarding



- How to determine to which LAN segment to forward a frame?
- Looks like a routing problem...

Self-learning, forwarding: example

- frame destination unknown: *flood*
- destination A location known: *selective send*

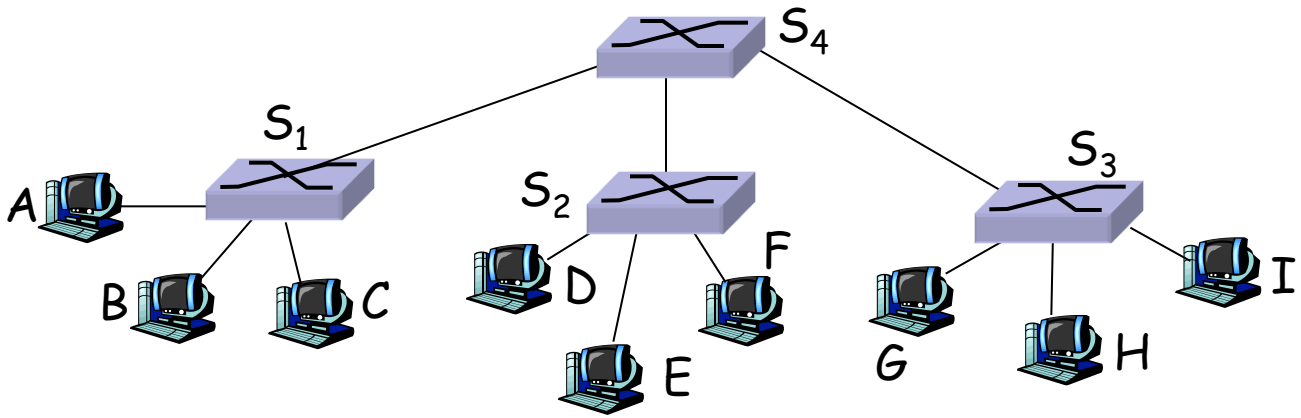


MAC addr	interface	TTL
A	1	60
A'	4	60

Switch table (initially empty)

Interconnecting switches

- switches can be connected together

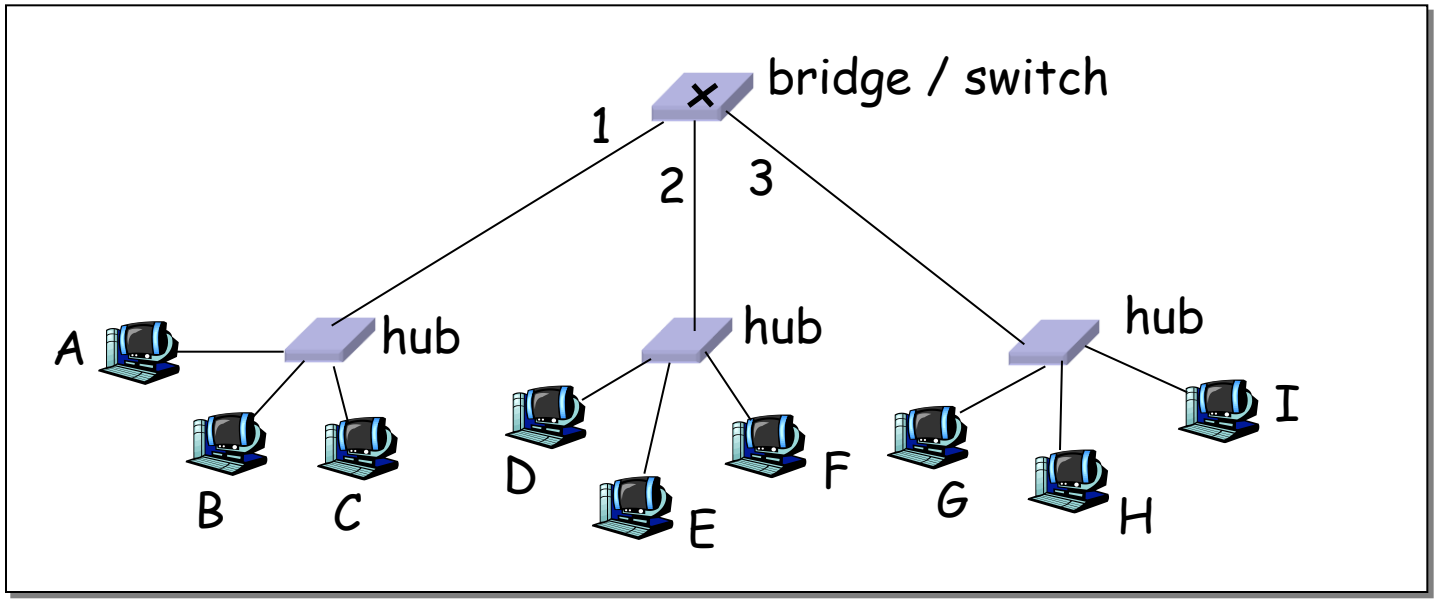


- *Q:* sending from A to G - how does S_1 know to forward frame destined to G via S_4 and S_3?
- *A:* self learning! (works exactly the same as in single-switch case!)

Switch example

address	interface
A	1
B	1
E	2
G	3

Suppose C sends frame to D

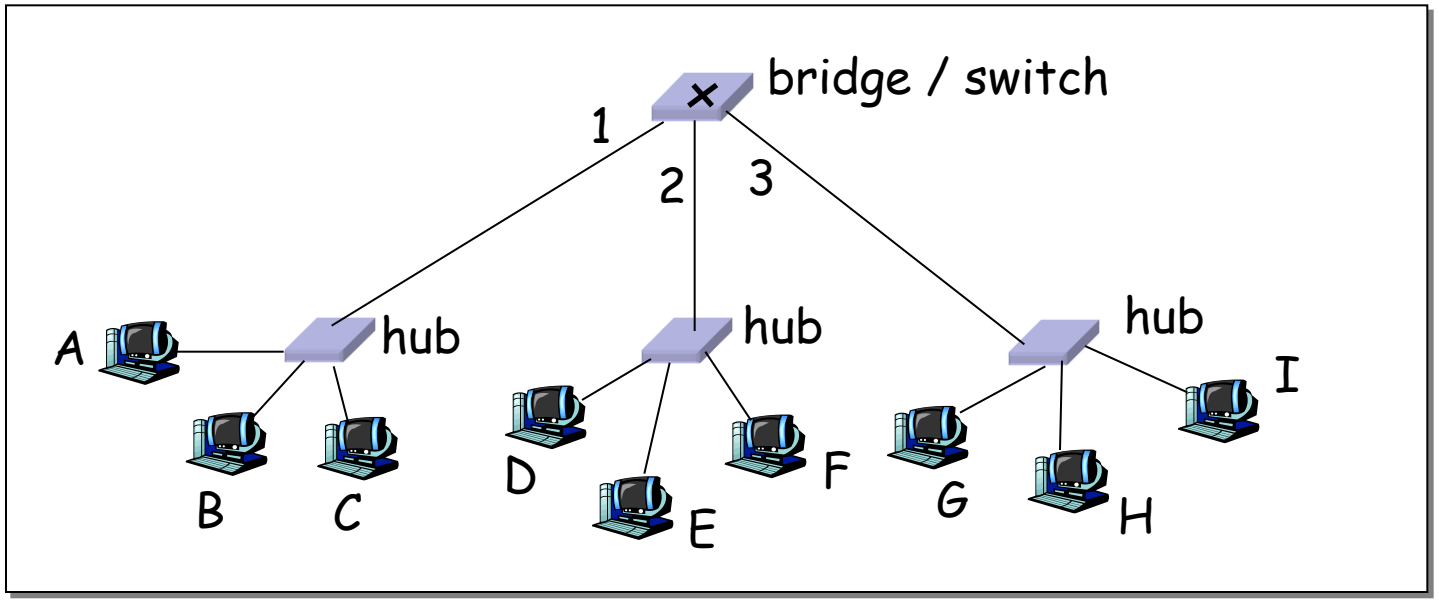


- Bridge receives frame from C
  - notes in bridge table that C is on interface 1
  - because D is not in table, switch forwards frame into interfaces 2 and 3
- frame received by D

Switch example

address	interface
A	1
B	1
E	2
G	3
C	1

Suppose D replies back with frame to C.

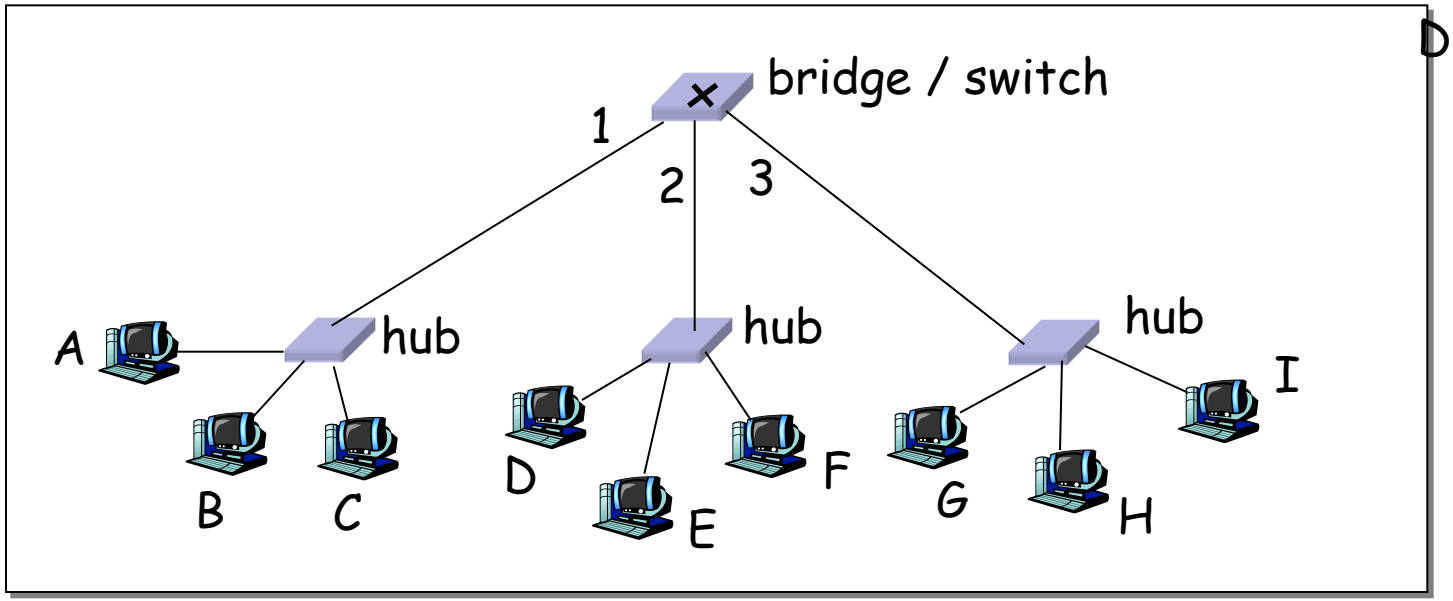


- Bridge receives frame from D
  - notes in bridge table that D is on interface 2
  - because C is in table, switch forwards frame only to interface 1
- frame received by C

Switch example

address	interface
A	1
B	1
E	2
G	3
C	1
D	2

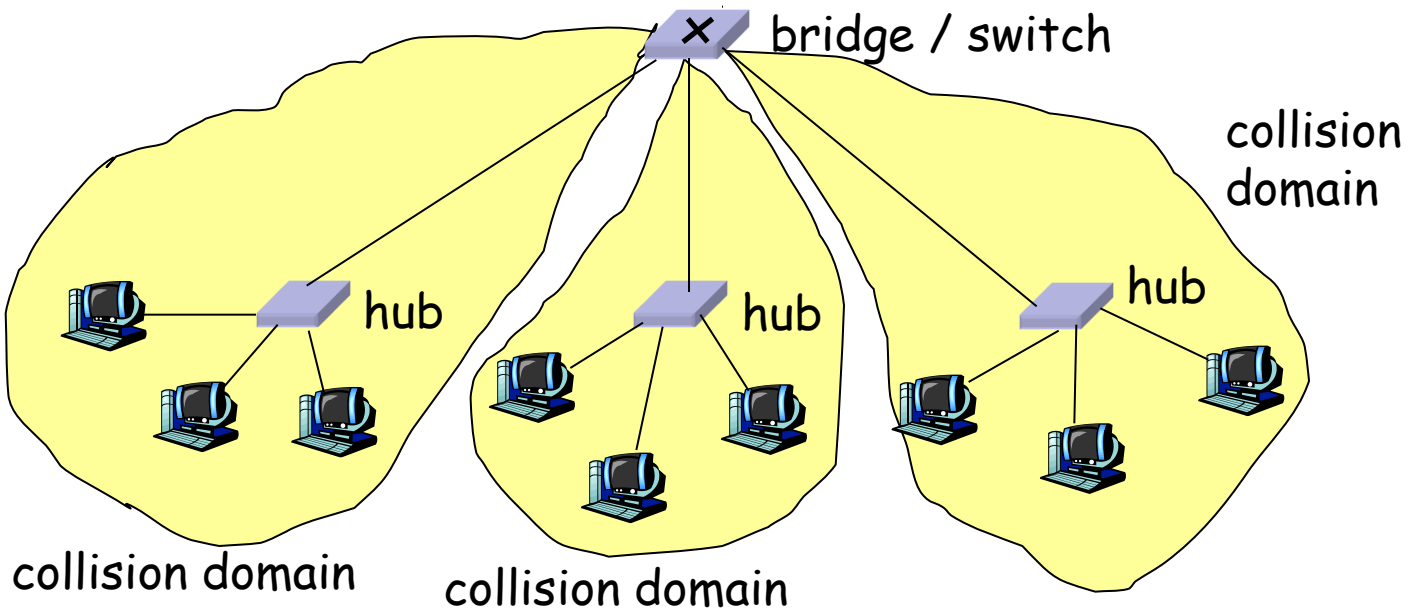
Suppose D replies back with frame to C.



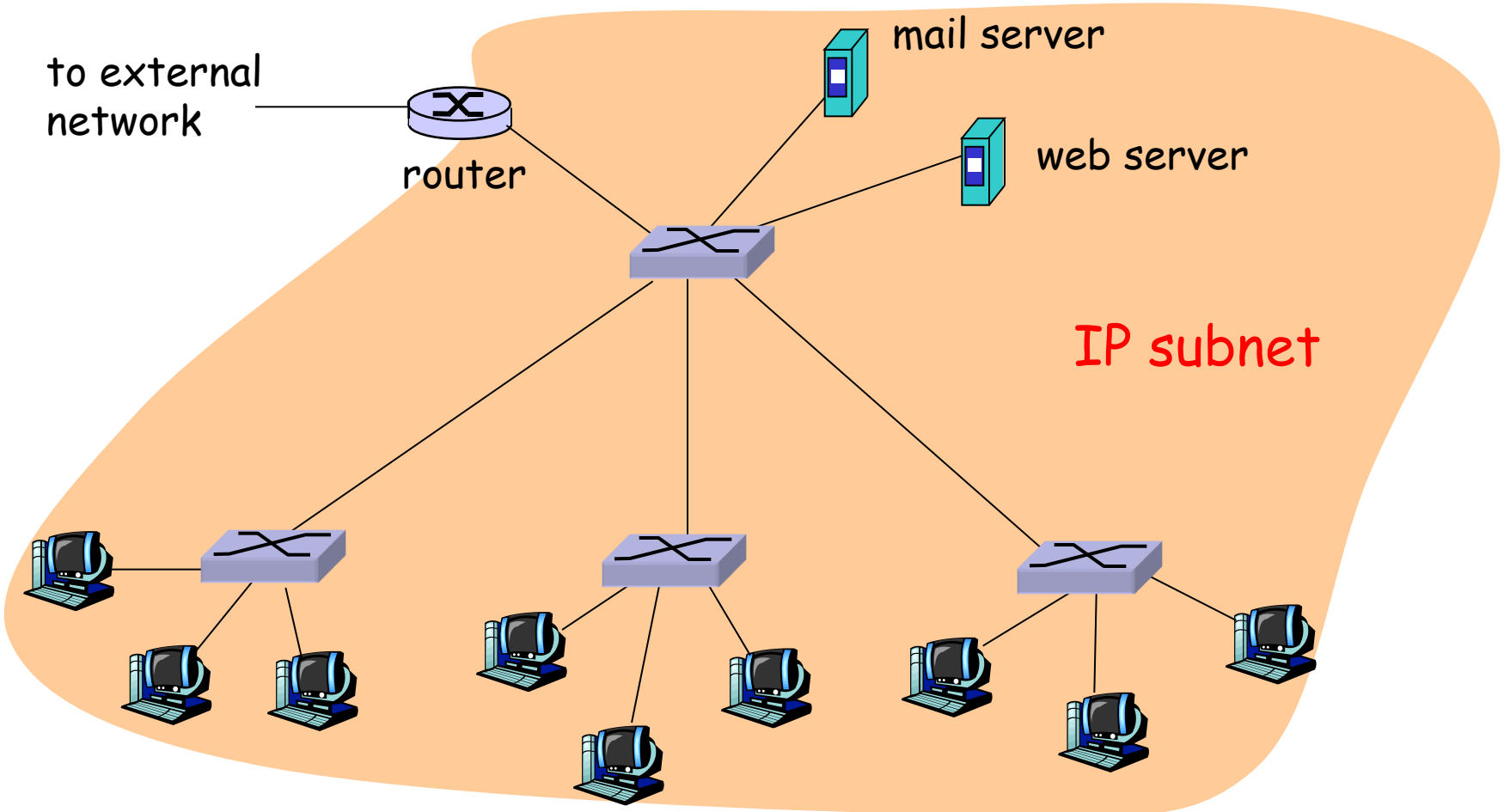
- Bridge receives frame from D
  - notes in bridge table that D is on interface 2
  - because C is in table, switch forwards frame only to interface 1
- frame received by C

Bridge: traffic isolation

- Bridge installation breaks subnet into LAN segments
- Bridge filters packets:
  - same-LAN-segment frames not usually forwarded onto other LAN segments
  - segments become separate collision domains



Institutional network



Routers vs. Bridges

- Bridge operation is simpler requiring less packet processing
- Bridge tables are self learning
- Router supports arbitrary topologies
- Router requires IP address configuration
- Router requires higher packet processing
- Bridges do well in small (few hundred hosts) while routers used in large networks (thousands of hosts)
- Unmanaged Bridges do not need a MAC and IP addresses

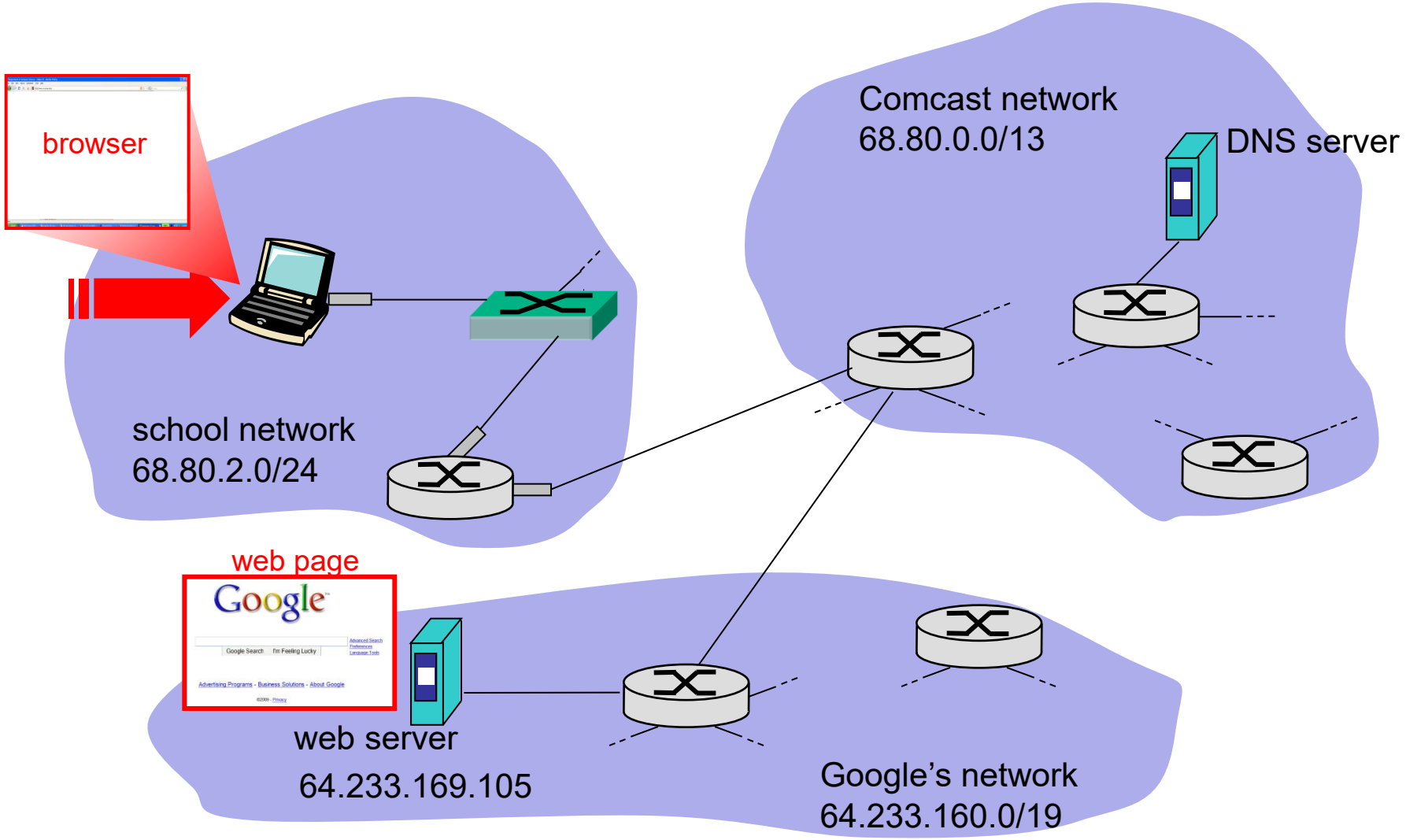
Today

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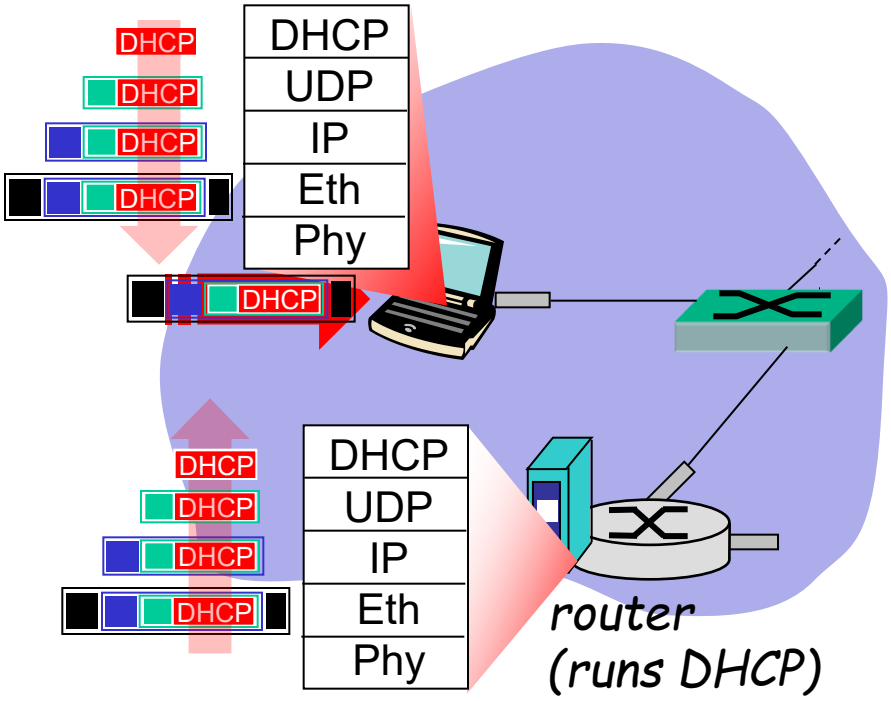
Synthesis: a day in the life of a web request

- Journey down protocol stack complete!
  - application, transport, network, link
- putting-it-all-together: synthesis!
  - *goal:* identify, review, understand protocols (at all layers) involved in seemingly simple scenario: requesting www page
  - *scenario:* student attaches laptop to campus network, requests/receives www.google.com

A day in the life: scenario

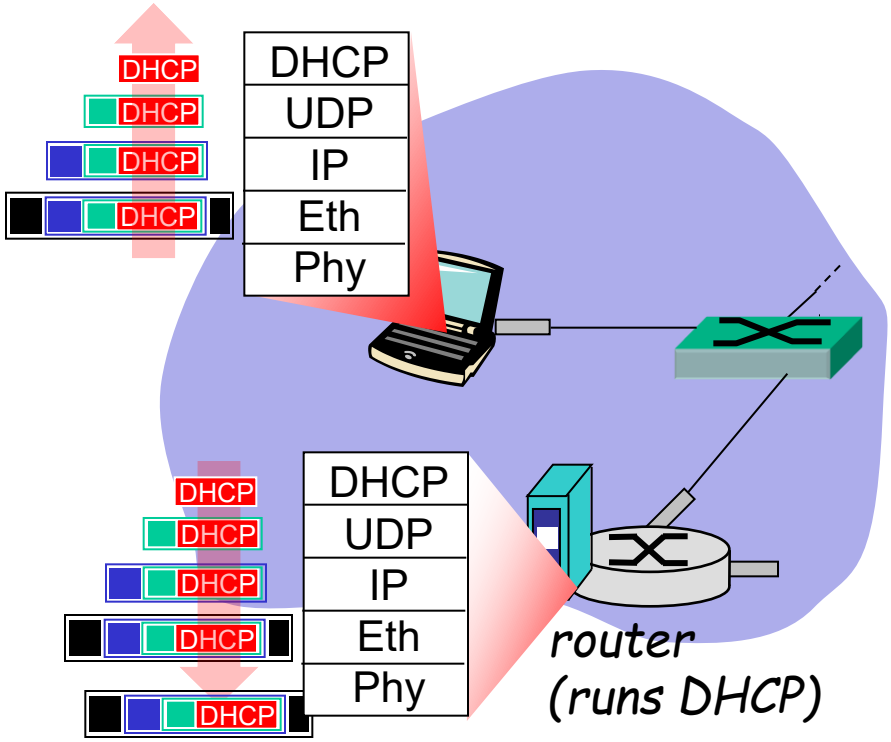


A day in the life… connecting to the Internet



- connecting laptop needs to get its own IP address, addr of first-hop router, addr of DNS server, musk: use ***DHCP***
- **DHCP discover *encapsulated*** in ***UDP***, encapsulated in ***IP***, encapsulated in ***802.1*** Ethernet
- Ethernet frame ***broadcast*** (dest: FFFFFFFFFFFF) on LAN, received at router running ***DHCP*** server
- Ethernet ***demux'ed*** to **IP** demux'ed, **UDP** demux'ed to **DHCP**

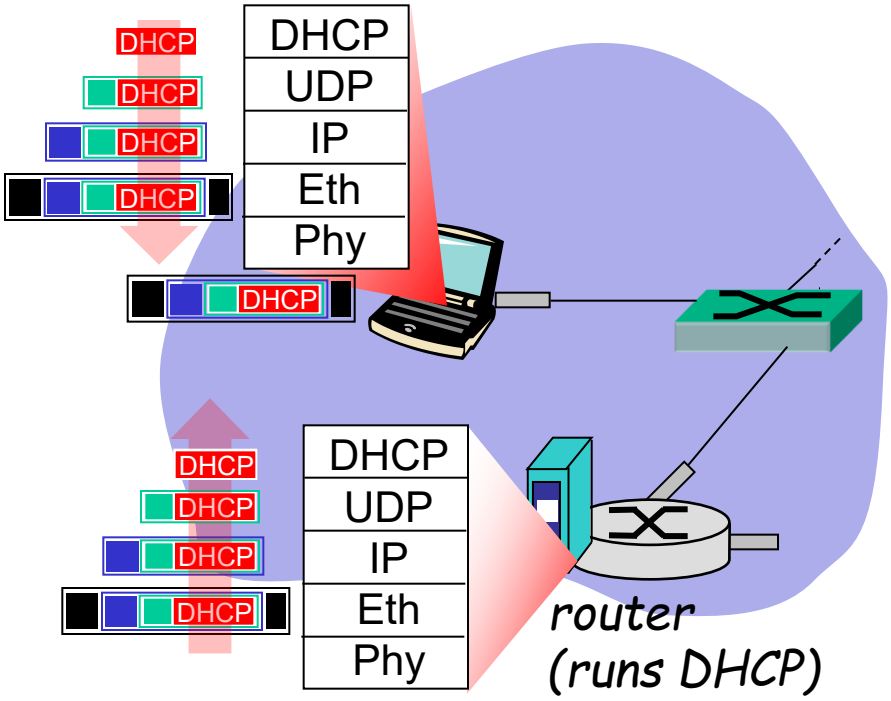
A day in the life… connecting to the Internet



- DHCP server formulates ***DHCP Offer*** containing (1) ofer for client's IP address,
- encapsulation at DHCP server, frame forwarded (***switch learning***) through LAN, demultiplexing at client
- DHCP client receives **DHCP Offer** reply

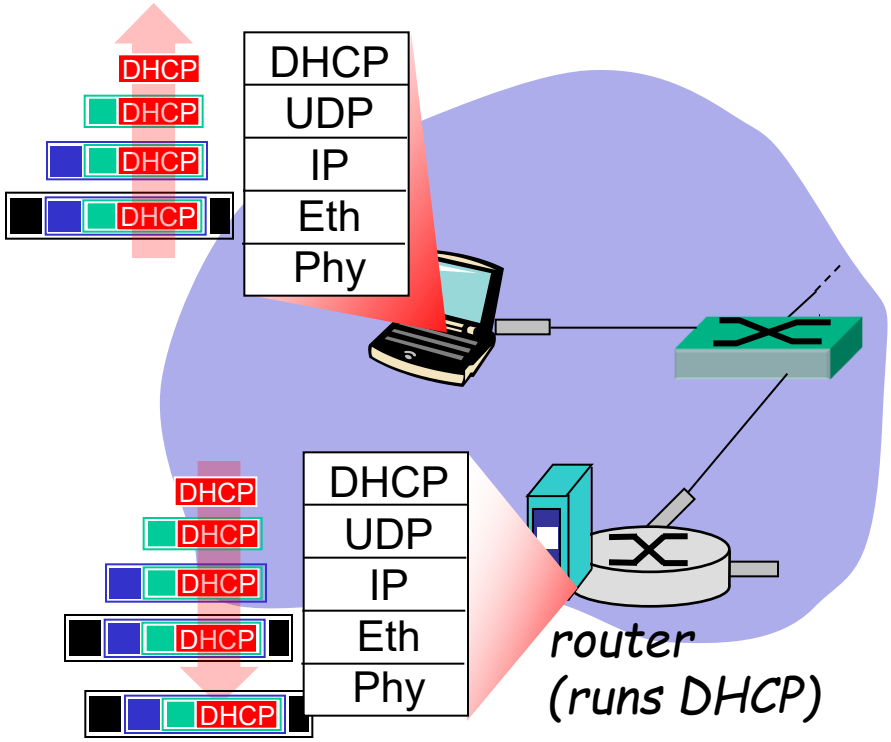
Client now (1) has IP address offer

A day in the life… connecting to the Internet



- DHCP request ***encapsulated*** in ***UDP***, encapsulated in ***IP***, encapsulated in ***802.1*** Ethernet
- Ethernet frame ***broadcast*** (dest: FFFFFFFFFFFF) on LAN, received at router running ***DHCP*** server
- Ethernet ***demux'ed*** to IP demux'ed, UDP demux'ed to DHCP

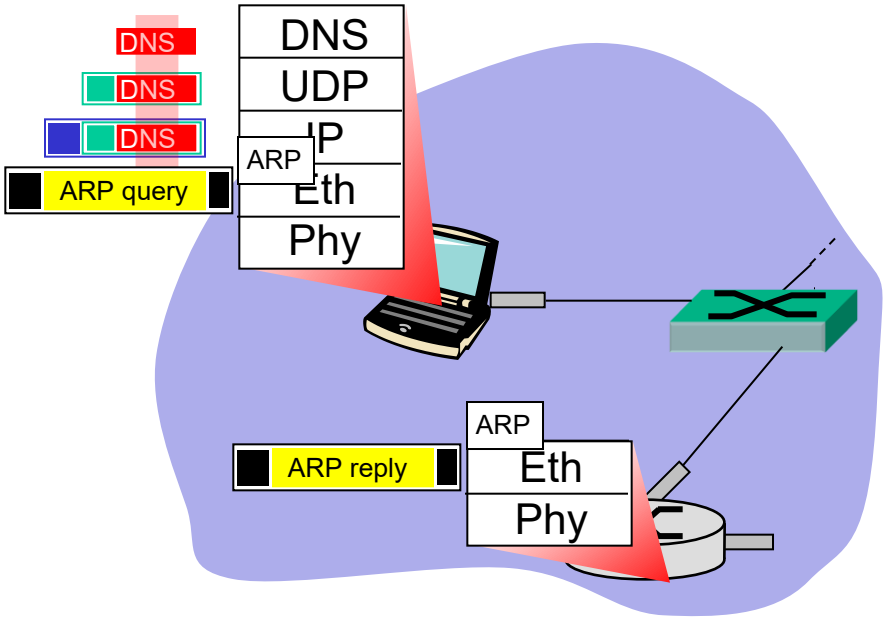
A day in the life… connecting to the Internet



- DHCP server formulates ***DHCP ACK*** containing (1) client's IP address, (2) IP address of first-hop router for client, (3) IP address of DNS server, (4) network mask.
- encapsulation at DHCP server, frame forwarded (***switch learning***) through LAN, demultiplexing at client
- DHCP client receives DHCP ACK reply

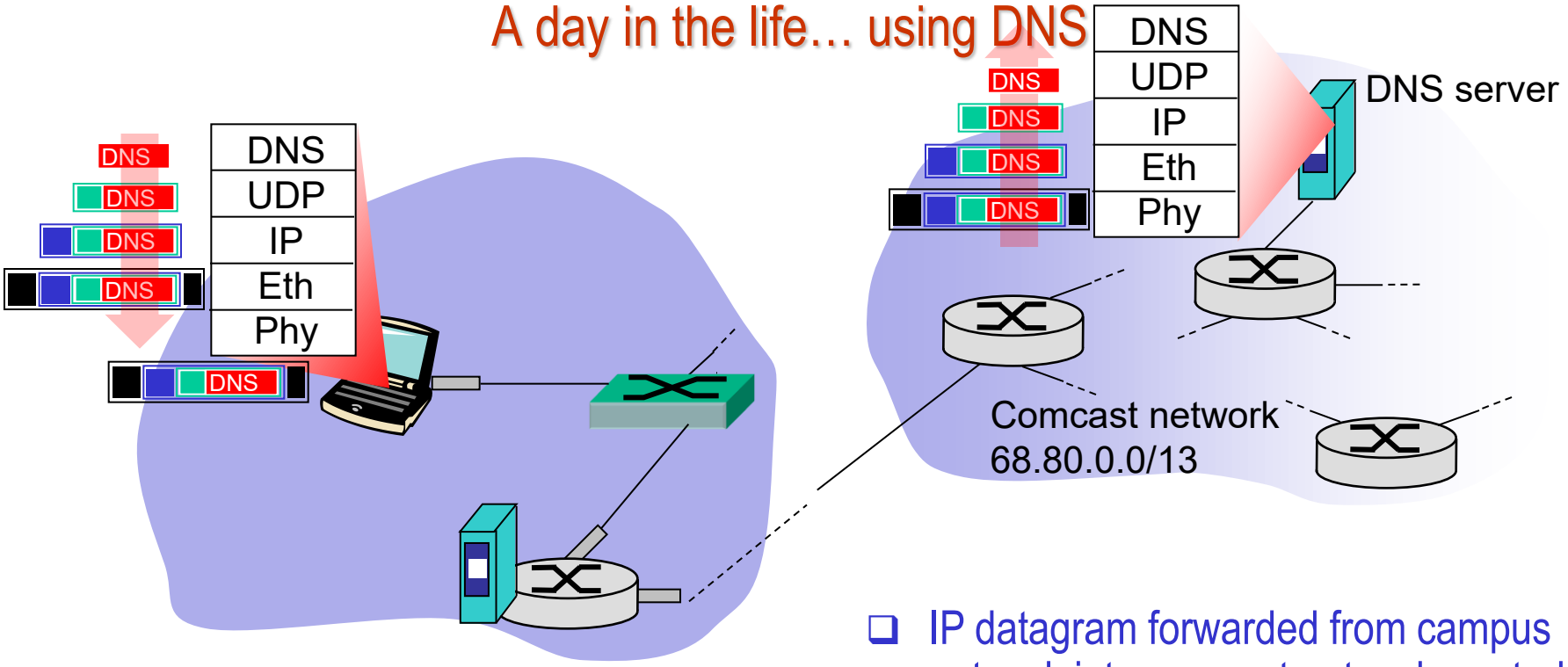
Client now (1) has IP address, (2) knows addr of DNS server, (3) IP address of its first-hop router, (4) mask.

A day in the life… ARP (before DNS, before HTTP)



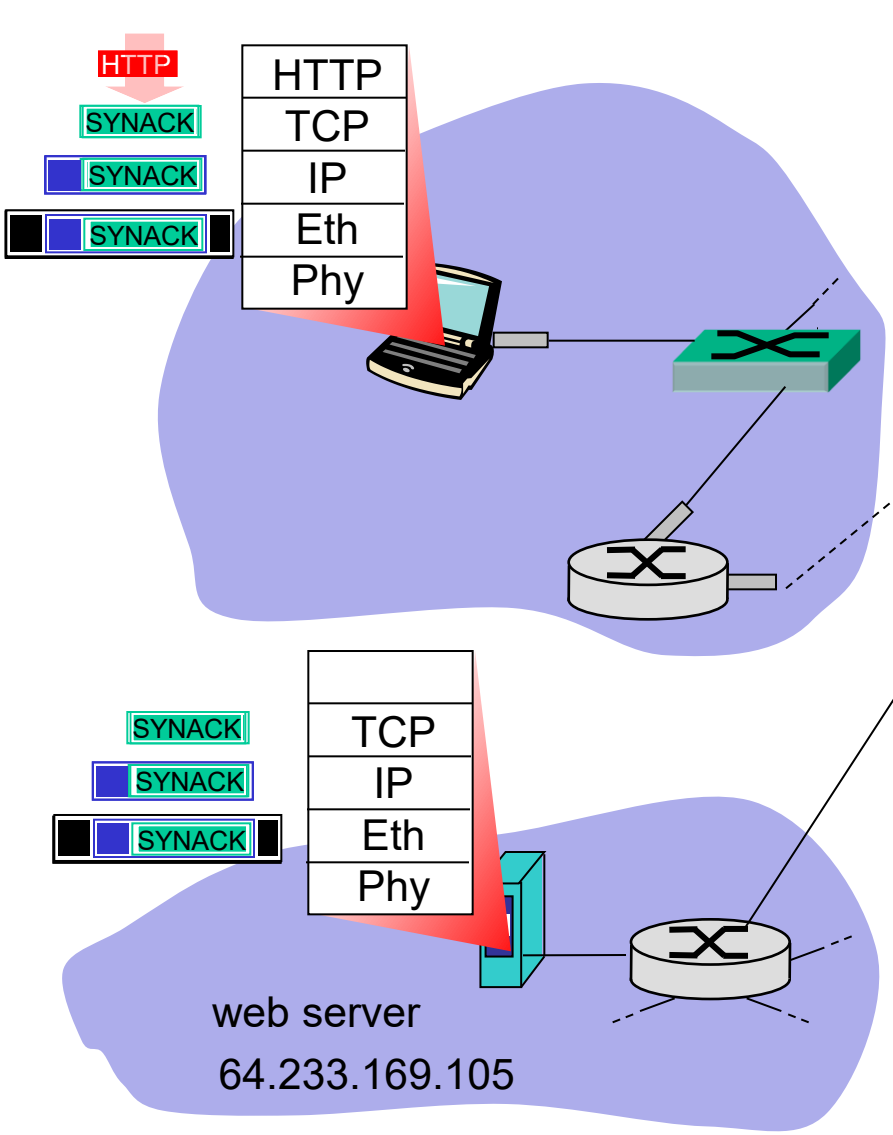
- before sending ***HTTP*** request, need IP address of www.google.com: ***DNS***
- DNS query created, encapsulated in UDP, encapsulated in IP, encasulated in Eth. In order to send frame to router, need MAC address of router interface: ***ARP***
  Network mask is used !!!
- ***ARP query*** broadcast, received by router, which replies with ***ARP reply*** giving MAC address of router interface
- client now knows MAC address of first hop router, so can now send frame containing DNS query

A day in the life… using DNS



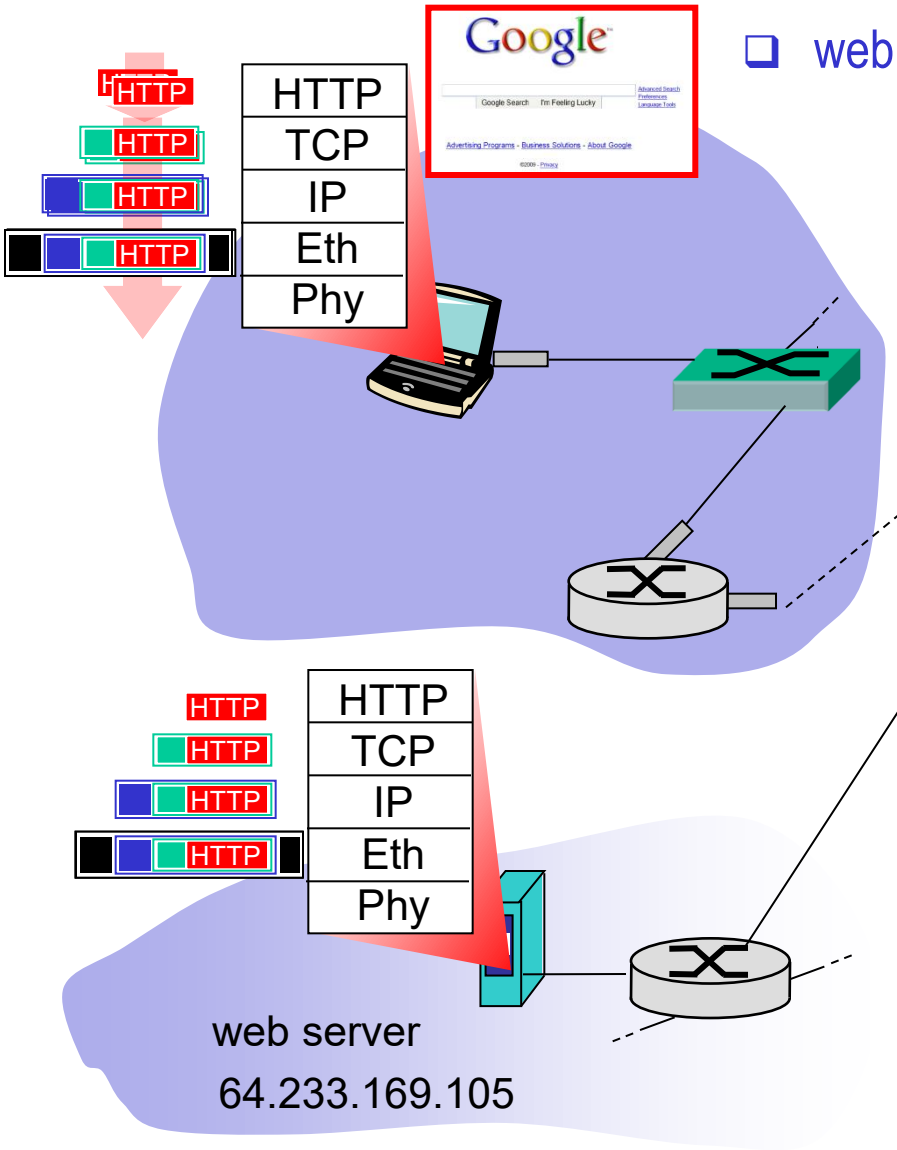
- IP datagram containing DNS query forwarded via LAN switch from client to 1st hop router
- IP datagram forwarded from campus network into comcast network, routed (tables created by ***RIP, OSPF, IGREP*** and/or ***BGP*** routing protocols) to DNS server
- demux'ed to DNS server
- DNS server replies to client with IP address of www.google.com

A day in the life… TCP connection carrying HTTP



- to send HTTP request, client first opens ***TCP socket*** to web server
- TCP ***SYN segment*** (step 1 in 3-way handshake) ***inter-domain routed*** to web server
- web server responds with ***TCP SYNACK*** (step 2 in 3-way handshake)
- TCP ***connection established!***

A day in the life… HTTP request/reply



- web page ***finally (!!!)*** displayed

- ***HTTP request*** sent into TCP socket
- IP datagram containing HTTP request routed to www.google.com
- web server responds with ***HTTP reply*** (containing web page)
- IP datgram containing HTTP reply routed back to client

Today

- ✓ Recap
- ✓ DHCP – some final details
- ✓ LAN technologies & collisions
  - ✓ T – connector bus
  - ✓ Hub
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Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

שאילתת ARP שנשלחת ב- collision domain מסוים לא תועבר ל-collision domain אחר.

1. נכון
2. לא נכון

An ARP query sent in one collision domain will not be passed into another collision domain

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

כאשר ***hub*** מקבל חבילה על אחד מקוי התקשורת, הוא תמיד שולח אותה על כל קוי התקשורת האחרים.

1. נכון
2. לא נכון

Whenever a hub receives a packet on one of its links, it always delivers it on all its other links

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

Hubs משפרים את הביצועי רשת לוקלית על ידי חלוקת הרשת למספר תחומי התנגשות.

1. נכון
2. לא נכון

Hubs improve the LAN performance by partitioning the network into a few collision domains.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

כאשר מתקינים bridge, יש צורך לעדכן בהתחלה את הכניסות בטבלת ה-bridge

1. נכון
2. לא נכון

When installing a bridge, it is first required to update its table entries.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

אם Bridge מקבל חבילה ואינו מזהה את כתובת ה- MAC של היעד,
הוא שולח את החבילה עם כתובת broadcast.

1. נכון
2. לא נכון

If a bridge receives a packet and does not recognize the destination MAC address, it forwards the packet with a broadcast address.

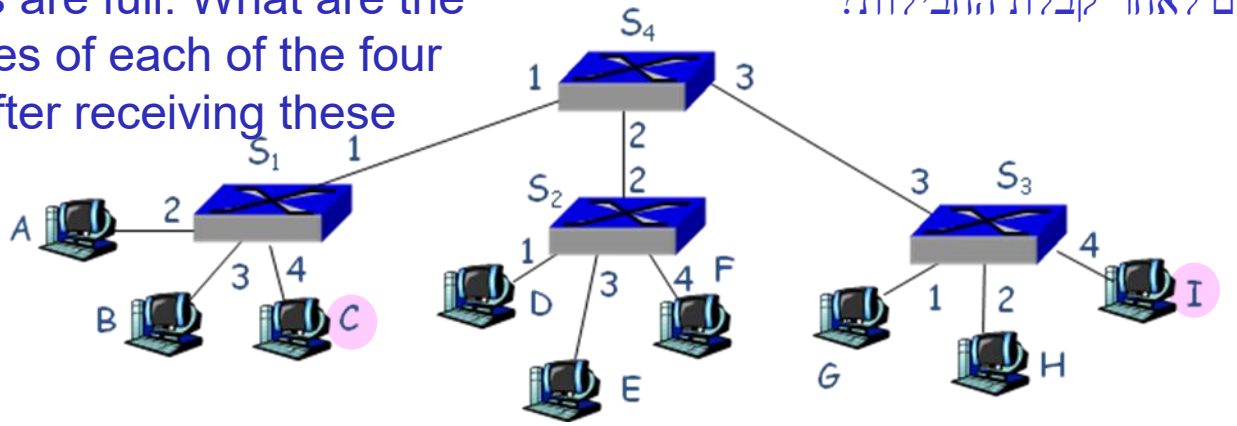
1. True
2. False

Question from an exam

שאלה מבחינה

Consider the figure with the 4 switches. Assume C sends an IP packet to I, and I upon receiving this packet sends a response IP package back to C. Assume that all ARP tables are full. What are the switch tables of each of the four switches after receiving these packets?

השאלה מתייחסת לרשת עם 4 המתגים (switches) בתמונה. הנח ש- C שולח חבילת IP ל- I , ו-I לאחר קבלת החבילה שולח בחזרה ל- C חבילת IP. נניח שטבלאות ARP מלאות. מה טבלאות המתגים של כל אחד מארבעת המתגים לאחר קבלת החבילות?



S1

address	interface
C	4
I	1

S2

address	interface
C	2

S3

address	interface
C	3
I	4

S4

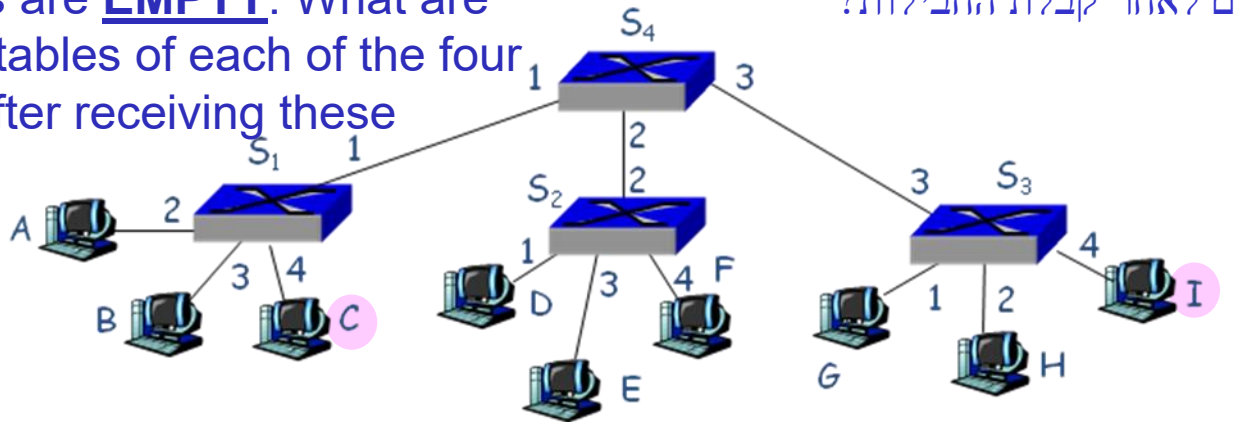
address	interface
C	1
I	3

Question from an exam

שאלה מבחינה

Consider the figure with the 4 switches. Assume C sends an IP packet to I, and I upon receiving this packet sends a response IP package back to C. Assume that all ARP tables are **EMPTY**. What are the switch tables of each of the four switches after receiving these packets?

השאלה מתייחסת לרשת עם 4 המתגים (switches) בתמונה. הנח ש- C שולח חבילת IP ל- I , ו-I לאחר קבלת החבילה שולח בחזרה ל- C חבילת IP. נניח שטבלאות ARP **ריקות**. מה טבלאות המתגים של כל אחד מארבעת המתגים לאחר קבלת החבילות?



S1

address	interface
C	4
I	1

S2

address	interface
C	2
I	2

S3

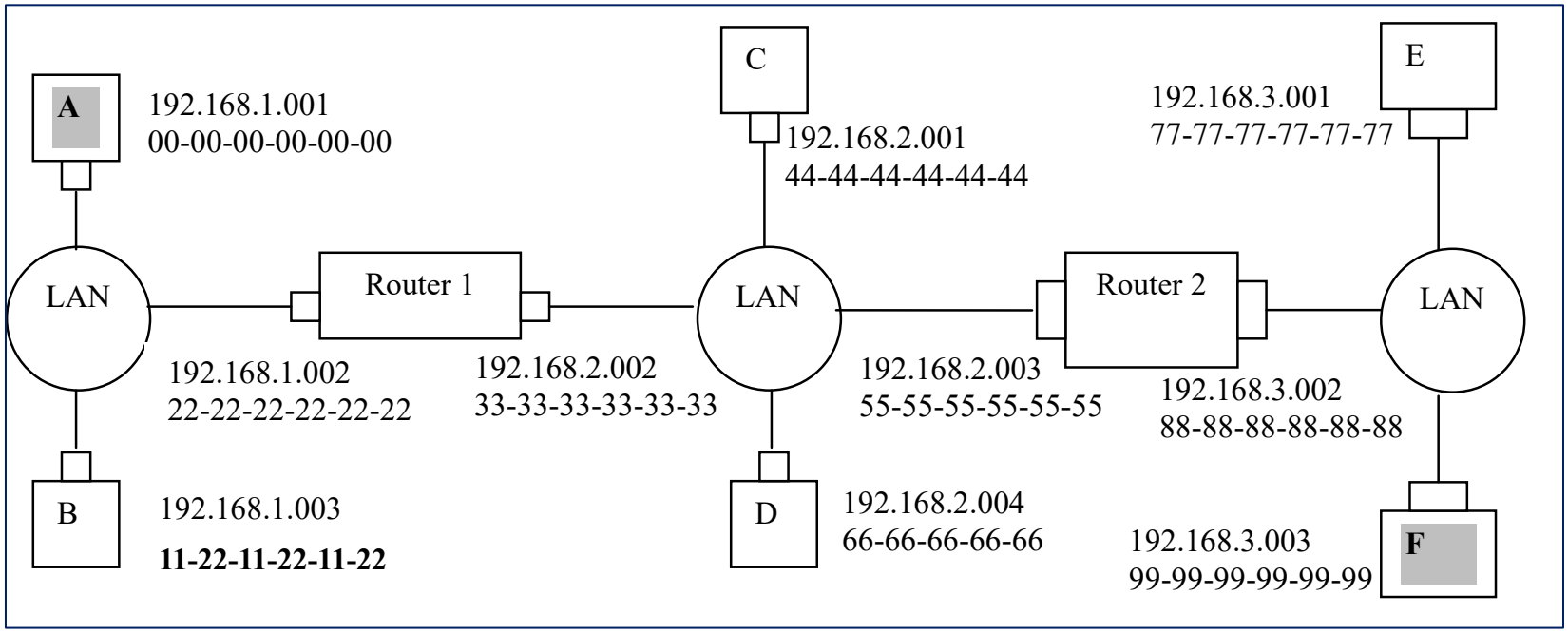
address	interface
C	3
I	4

S4

address	interface
C	1
I	3

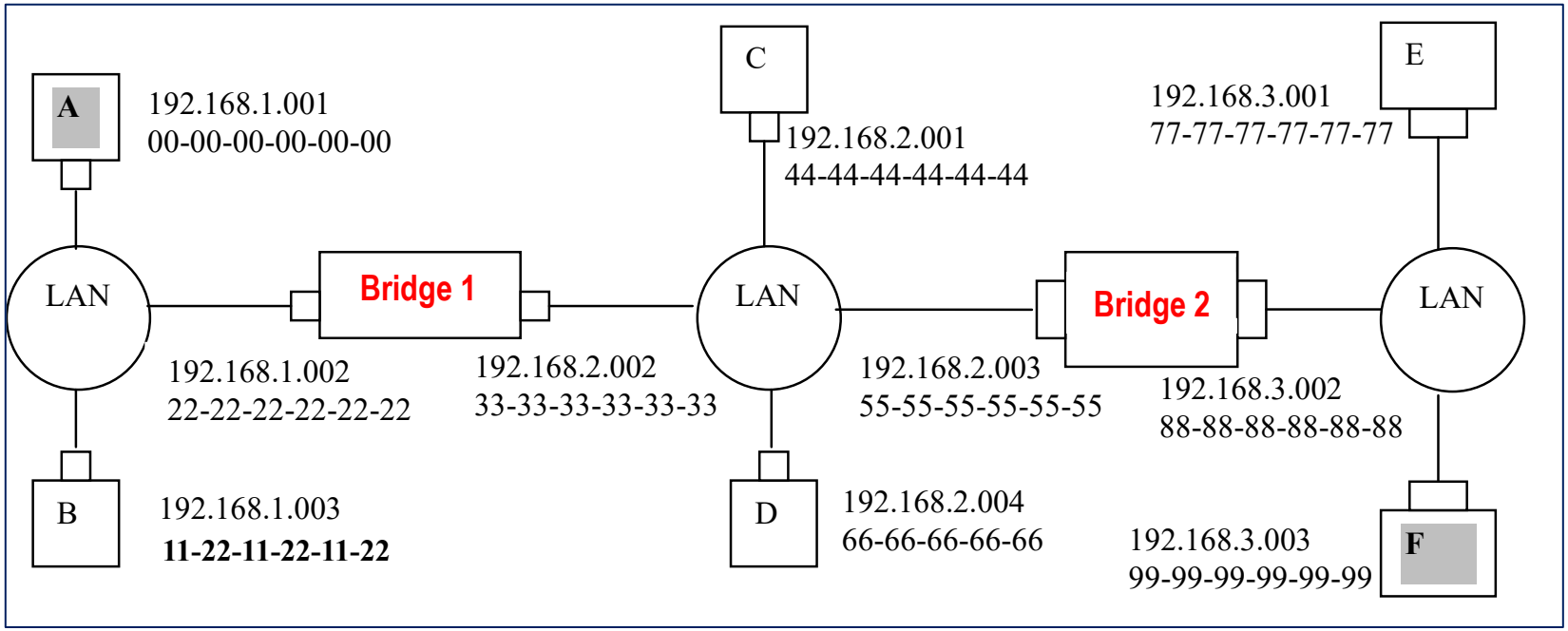
שאלה מהרצאה קודמת

השאלה מתייחסת לרשת בתמונה בהמשך. הנח שיש לשלוח חבילה אחת ממחשב A ל- מחשב F.
הנח שכל טבלאות ה- ARP ריקות וכל טבלאות הניתוב מעודכנות. רשום בטבלה בהמשך את כל החבילות שנשלחות עד שהחבילה מגיעה ליעד. ברישום בטבלה שמור על סדר השליחה.
הערה: מספר השורות בטבלה לא בהכרח משקף את המספר הכולל של חבילות שנשלחו.



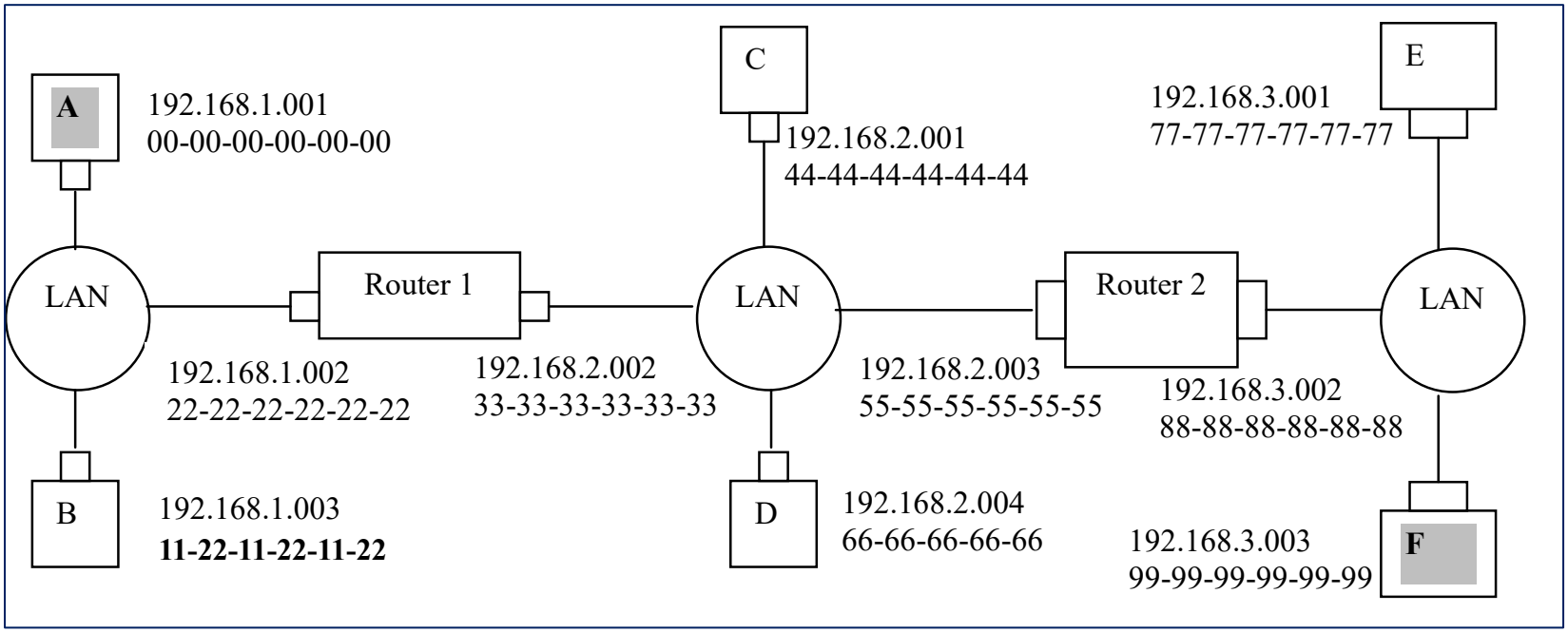
שאלה מבחינה

השאלה מתייחסת לרשת בתמונה בהמשך. הנח שיש לשלוח חבילה אחת ממחשב A ל- מחשב F.
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Question from the previous lecture

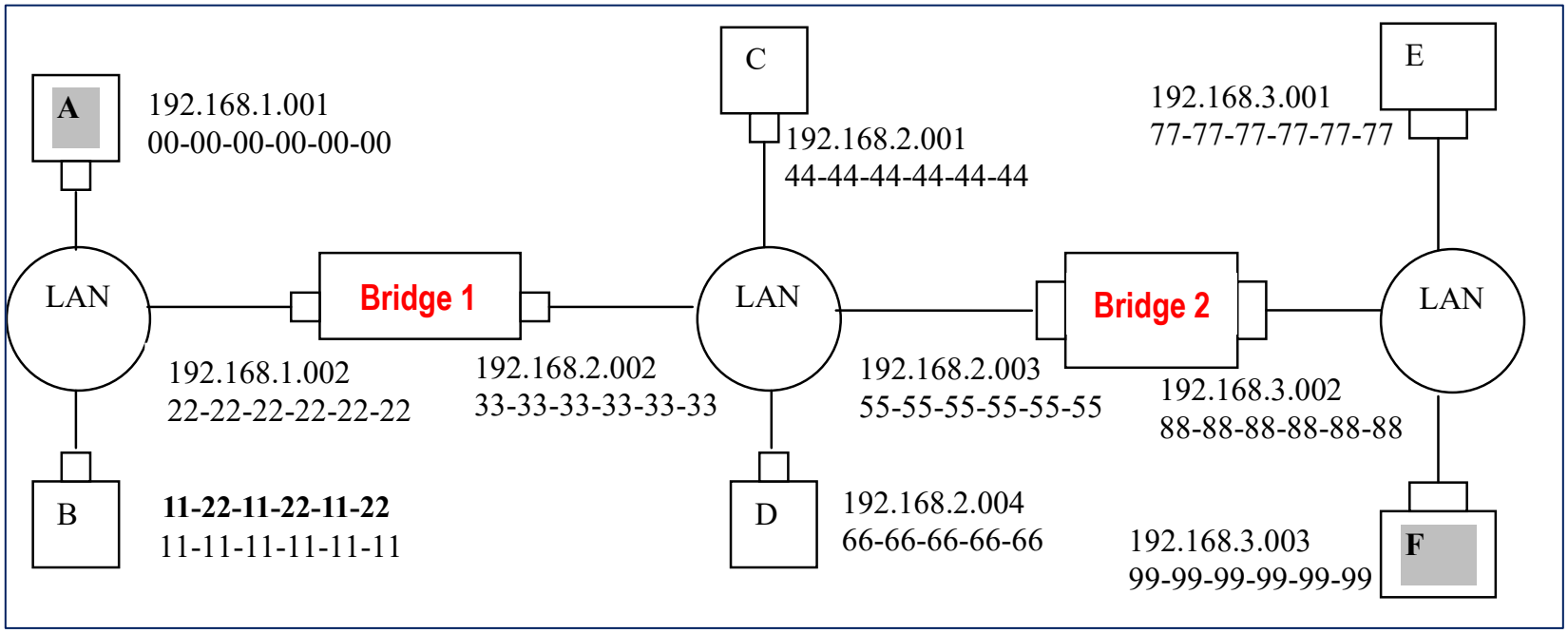
Consider the network presented in the next figure. Assume there is one packet to be sent from host A to host F. Assume all ARP tables are empty, but all forwarding tables are updated. Fill in the (next slide) table the header details of all the packets which are sent until our desired packet finally reaches the destination. Write the records in their correct order. Note: the number of rows in the table does not necessarily represent the number of packets which are sent.

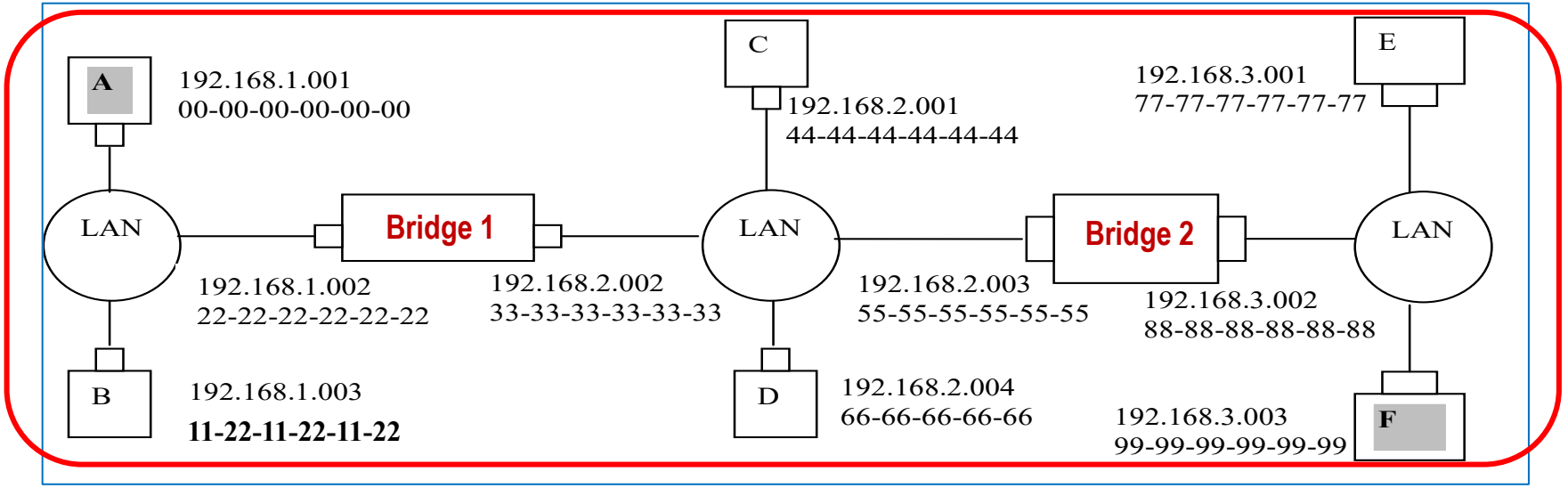


Question from an exam

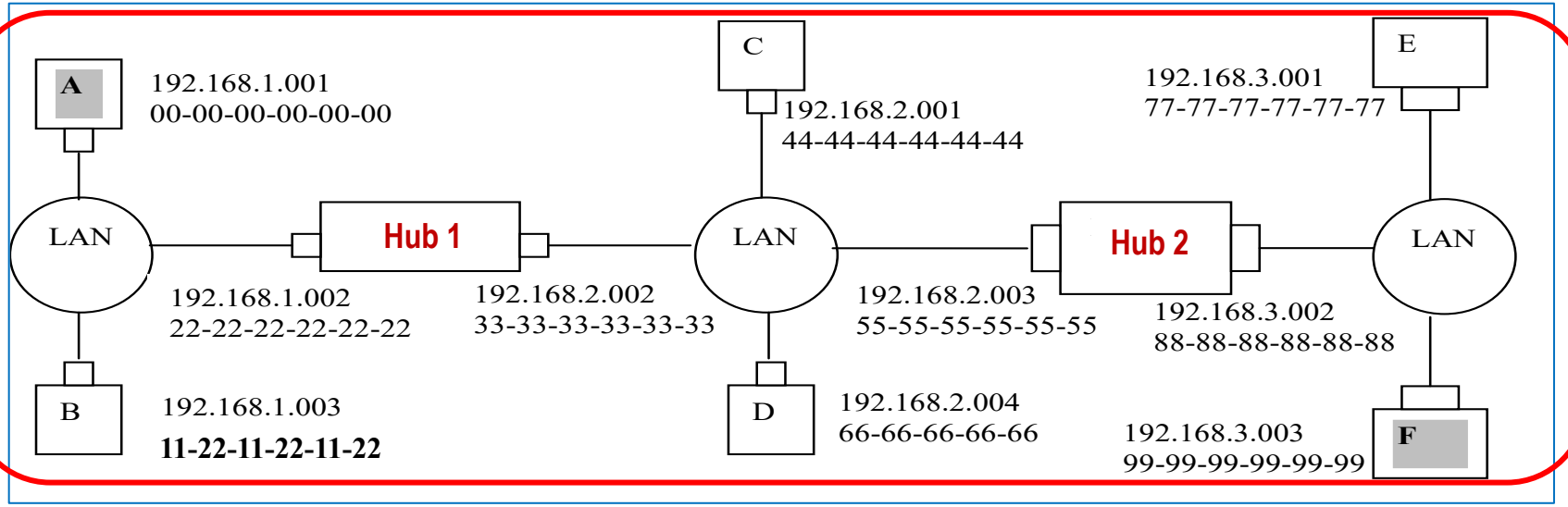
שאלה מבחינה

Consider the network shown in the figure below. Assume now both routers are replaced with two bridges. Assume a packet should be sent from host A to host F. Assume all ARP tables are empty, and all the routing tables are updated. Fill in the following table all the packets which are sent until the packet from A reaches the destination. Keep the order of the packets in the table similar to the order of the time in which they are sent. The number of rows in the table does not necessarily represent the number of packets which are sent.

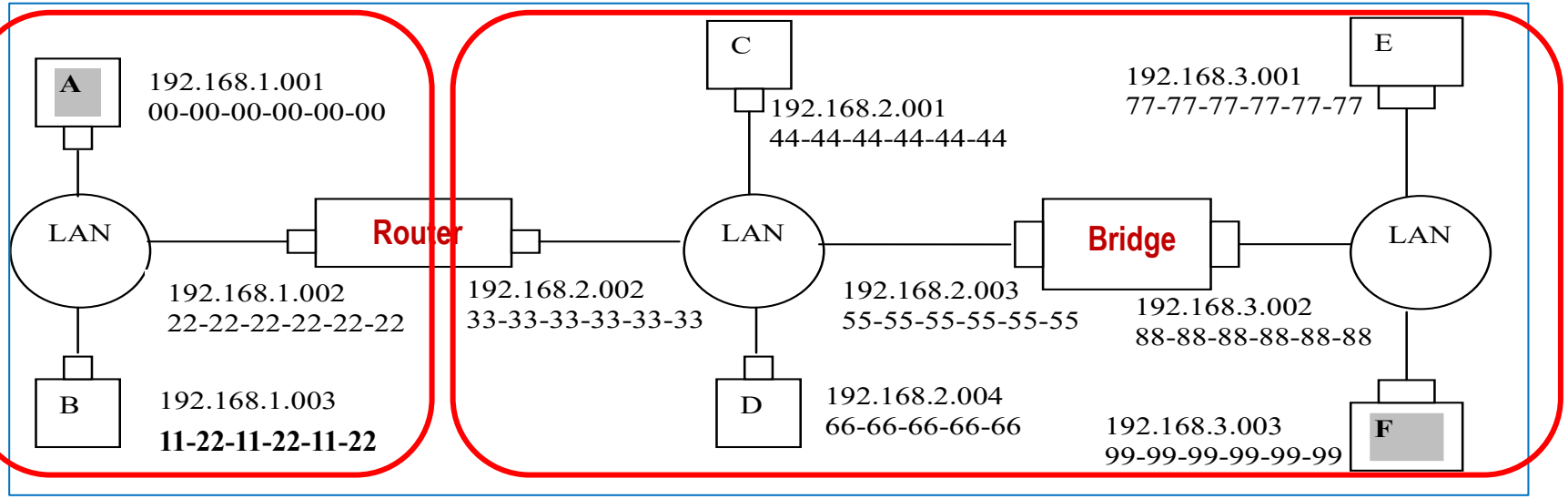




Frame		MAC Address		IP Address	
#	Type IP/ARP	source	destination	source	destination
1	ARP	A 00-00-00-00-00-00	FF-FF-FF-FF-FF-FF		
2	ARP	F 99-99-99-99-99-99	A 00-00-00-00-00-00		
3	IP	A 00-00-00-00-00-00	F 99-99-99-99-99-99	A	F
4					
5					
6					
7					
8					
9					



Frame		MAC Address		IP Address	
#	Type IP/ARP	source	destination	source	destination
1	ARP	A 00-00-00-00-00-00	FF-FF-FF-FF-FF-FF		
2	ARP	F 99-99-99-99-99-99	A 00-00-00-00-00-00		
3	IP	A 00-00-00-00-00-00	F 99-99-99-99-99-99	A	F
4					
5					
6					
7					
8					
9					



Frame		MAC Address		IP Address	
#	Type IP/ARP	source	destination	source	destination
1	ARP	A 00-00-00-00-00-00	FF-FF-FF-FF-FF-FF		
2	ARP	R1-left 22-22-22-22-22-22	A 00-00-00-00-00-00		
3	IP	A 00-00-00-00-00-00	R1-left 22-22-22-22-22-22	A	F
4	ARP	R1-right 33-33-33-33-33-33	FF-FF-FF-FF-FF-FF		
5	ARP	F 99-99-99-99-99-99	R1-right 33-33-33-33-33-33		
6	IP	R1-right 33-33-33-33-33-33	F 99-99-99-99-99-99	A	F
7					
8					
9					

Today

- ✓ Recap
- ✓ DHCP – some final details
- ✓ LAN technologies & collisions
  - ✓ T – connector bus
  - ✓ Hub
  - ✓ Bridge & Switch
- ✓ A life of a web request
- ✓ Questions